Teaching research data management skills using resources and scenarios based on real data

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Our experiences

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Overall goals of good data practices

- Complete and self-explanatory data collections for that users can understand and use
- Do no harm

Challenges

- Domain-specific for social sciences: surveys, interviews – may not apply to other sciences on the surface, but principles do apply, e.g. labelling, documenting, enhancing, disclosure review
- Level of experience participants have with data and software – make exercises software agnostic (e.g. not R, Stata), else time goes on learning the software
- Sophistication of user – do they know what data files are, how data are structured, what labelling is?
- Balance messy vs simplistic data exercises

What works

- Single data collection that can be used for teaching variety of topics
- Materials contributed in advance by participants: DMP, consent forms, ethical challenges/dilemmas, research summary … small items that can be shared with participants, or worked into scenarios
- Guided step-by-step exercises as warm-up exercises
- Mixed audiences, senior researchers, challenges that cannot be resolved
- Take existing data collection and mess it up

Know your audience

- Data professionals – training in data curation, review,….
- Researchers – training in good data practices
- Junior vs senior, e.g. postgrads

Topics to cover

- Data review: what is missing? Out of range codes, what doesn’t make sense,
  - Quality of data
  - Documentation, labelling, metadata
- Data > people, so ethical / legal implications:
  - Disclosure review
  - Anonymisation
  - Consent in qualitative research (UK)
- How much time does your audience have
- Lots of other RDM aspects can be relevant too

Training methods

- Hands-on: exercises, scenario discussions, demos (e.g. encryption, checksums, backups)
- Real data / real research

Developing training exercises - UKDA

- Health and Social Consequences of the Foot and Mouth Disease Epidemic in North Cumbria, 2001-2003
  - interview recordings, transcripts and diaries
  - Safeguarded, embargoed and permission access
- Conflicts and Violence in Prison, 1998-2000
  - interviews, also coded responses
  - permission access
- Polish and Lithuanian Workers: Opportunities and Challenges for Trade Unions, 2004-2006
  - Survey, incl. freetext responses
  - safeguarded
- Life on antiretroviral therapy: People's adaptive coping and adjustment to living with HIV as a chronic condition in Wakiso District, Uganda
  - Interviews, ethnographic observations, questionnaire survey
  - safeguarded

Why suitable for RDM exercises

- Research topics that many researchers would understand
- Methodologies that most researchers understand
- Mixed methods: survey and interview
- Sensitive topics, so legal and ethical aspects are important aspects to showcase
- Manageable size data collections
- Well documented
- Some messiness in the data

How do I develop these into exercises

- Leave real messiness in, e.g. tracked changes in documents, poor labelling, inconsistent formatting, disclosive information in freetext variables, poor file names,…
- Insert new messiness e.g. fake names, disclosive variables, disclosive info in file properties, …. based on what we typically see in data deposits
- Reduce size of dataset, e.g. up to 20 variables, up to 10 responses for survey file; extracts of interview transcript (few pages); amended codebook
- Ensure data can be used openly in workshop, so scramble data around or apply pseudonyms
- Get permission of depositor
- Write instructions, research summary, dataset overview
- Write discussions / responses, marke-up versions (e.g. anonymisation)

Example: context

Russell, Steven (2015). *Life on antiretroviral therapy: People's adaptive coping and adjustment to living with HIV as a chronic condition in Wakiso District, Uganda.* [Data Collection]. Colchester, UK Data Archive. 10.5255/UKDA-SN-851094

- The research in Entebbe, Uganda, analyses the experiences of people living with HIV following access to life-saving antiretroviral therapy (ART). The study aims to understand how people have responded to a new chance at life, what factors enable people to adjust to living with HIV as a chronic condition, and what support measures affect this adjustment
- Adjustment to a new life on ART poses medical, social and economic challenges, especially in settings of poverty. People must take treatment for the rest of their lives, and they are usually recovering after a period of serious illness and disruption to their social and economic lives.
- Three ART delivery sites with different modes of delivery and support were compared:
  - HIV clinic in Entebbe hospital
  - 3 government health centres with referral links to Entebbe hospital
  - The Aids Support Organisation (TASO), Entebbe branch
- A quasi-experimental research design was used to compare people affected and unaffected by HIV and ART.

Example: data collection

- Life-history and illness narratives from 38 participants, recruited from the three study sites: notes taken during interview and narrative written out afterwards
- Semi-structured interview from same 38 participants examining key issues of HIV self-management: interview taped, transcribed verbatim and translated to English
- Questionnaire with 263 people living with HIV and 160 community control group participants: questionnaire completed by trained enumerators
- The survey questionnaire measured Illness Perceptions (IPQ brief standard), Quality of Life (WHOQOL-BREF standard), mood (Hopkins symptom checklist), mental adjustment to HIV (MAHIVS scale) and socio-economic status. In addition, participants provided information of their age, gender, household size, years in education, religion and marital status.
- Ethical approval for the study was obtained from the Uganda Virus Research Institute and the University of East Anglia, UK. Overall approval was granted by the Uganda National Council for Science and Technology.

Example: data and documentation files

InterviewExtracts	08/03/2016 11:12	File folder	
2012_01_18__Interview_3__Final_Copy__CASES__gone_to_printer	15/07/2015 07:01	Adobe Acrobat D...	985 KB
Background_description_for_the_qualitative_data	15/07/2015 07:01	Microsoft Word D...	21 KB
Background_description_for_the_quantitative_data	15/07/2015 10:38	Microsoft Word D...	35 KB
FINAL_Info_Sheet__26_consent_forms_Life_on_ART (1)	15/07/2015 07:01	Microsoft Word 9...	61 KB
HIVUganda_codebook_partial	28/07/2015 08:02	Adobe Acrobat D...	398 KB
HIVUganda_infosheet	14/07/2015 19:16	Microsoft Word 9...	61 KB
HIVUganda_surveydata_extract	15/07/2015 12:55	Microsoft Excel W...	15 KB
journal.pone.0105154	15/07/2015 12:31	Adobe Acrobat D...	202 KB
ResearchSummary_HIVUganda	15/07/2015 07:02	Microsoft Word D...	20 KB

Extract_interviewnarrative	28/07/2015 08:02	Microsoft Word 9...	50 KB
Extract_interviewobservations	28/07/2015 08:02	Microsoft Word 9...	40 KB
Extract_interviewtranscript	28/07/2015 08:02	Microsoft Word 9...	48 KB

Example: transcript extract

Extract: Narrative based on notes from first interview

ID NO: 5024

SN: Jackie

RESPONDENT INITIALS: AK

SEX: FEMALE

MARITAL STATUS: Single

AGE: 61

VILLAGE: Lugonjo, Entebbe

Summary data:

Age at first interview	61
Marital status	Single (separated)
Main activity	Cleaning/gardening
Occupation	Support staff at Entebbe Municipal Council
CD4 on diagnosis	277
CD4 now	192 (last tested in 2008)
AIDS-related symptoms now	None
ART source	TASO
ART regimen	1st
Adherence	Yes
Intimate partner	None
Safe sex	N/A
Other useful characteristics	Her sources of income are inadequate and sometimes she has to take ARVs on an empty stomach
E.g. Income sources	
HH characteristics	Says she lives alone in a rented house

Interview1(a)

INTERVIEW DATE: 26/05/11

DURATION: 10:30hrs-12:00 hrs

INTERVIEWER: SN

LOCATION: Uganda Virus Research Institute clinic-counseling room

Narrative

AK said thats he was sixty one years old but did not know her date of birth. However she was sure that in 1965, the year in which she gave birth to her first child she was fifteen years old. She is a Mutooro by tribe and was born in Gomba which at the time was part of Mpigi district although her ancestral home is in Kahunge in Kamwenge District (western Uganda).

She was born in a family of twelve children but nine of them died before the age of two and only three grew into adults. AK does not know why her siblings died as babies. Of the three that made it to adulthood, two children including AK were girls and there was one boy. Today, AK is the only living child of her parents as her two siblings have since passed on. The boy died in 1974 at the age of seventeen and AK said that he died after vomiting and having diarrhea for three days. Then her sister died in 2010 from ulcers.

I then asked AK how she felt about the deaths of all her siblings and she told me, *"nokoloki? Kotonda yankumye era obulwadde bweyampa bwebuli ntwala yolimanya"* meaning, what can I do? God has kept me alive and if I eventually die from the disease that he gave me, it will be up to him.

AK's father was a farmer, he reared cattle and grew cotton on a large scale and this was his source of income. Her mother was a house wife but she died during child delivery when AK was still a very young girl. AK told me that her parents managed to take very good care of the family despite their little sources of income, *"okulabirira okwedda kwolimu kki? Osala okabuzi era nolima ne mmere yo eyokulya"* meaning, what was hard about taking care of people those days? You would slaughter a goat and cultivate your food to eat.

At the time AK was born, her father was an old man and even though he loved her so much she blames him for not having educated her, *"taata yali atwagala naye kyeyakola ekibi teyansomesa"* meaning that father loved us but the bad thing he did was not educating me.

I then asked why her father did not educate her and she told me, *" taata yali aneenya maama olwa abaana abaafa era naa amuyita kisirani era namulekenera"* meaning that father blamed mother for all the dead children and used to call her cursed so he abandoned her. She went on and said, *" nazalibwa mu naaku nyingi nya musawa"* meaning that I was born amidst a lot of sorrow.

face). I then asked her to explain further and she told me, *"amusaija yalaluka nafuna abakazi abalala naye muiia wange nampisa bubbi kuba yali tazala era natutabula"* meaning that the man became wild and he started getting involved with other women and my co-wife mistreated me because she was not able to bear children so she came between us. *" twali bota ffembi era amusaija yali anjagala naye mu lukumi mu lwenda okaga mu musanyu namuvira"* meaning that we were both young and the man loved but in 1967 I left him.

After AK left W she went back to Kahunge in Kamwenge district and towards the end of 1967 met a man

Example: transcript extract

ID NO: 73

Extract: transcript from interview recording

ID NO: 73
JK: Derrick
MAIN STUDY INTERVIEW: LIVING ON ART STUDY
RESPONDENT INITIALS: MP
SEX: MALE
MARITAL STATUS: MARRIED
AGE: 38 YEARS
VILLAGE: SSISA, NAKAWUKA
INTERVIEWER: JK

Age at first interview	38 years
Marital status	Married
Main activity	Excavating
Occupation	Works at the Stone quarry
CD4 on diagnosis	Not known: after start of ART, 35
CD4 now	Not known
AIDS-related symptoms now	Dark nails
ART source	Nakawuka
ART regimen	1st
Adherence	Yes
Intimate partner	Yes
Safe sex	No
Other useful characteristics	
Eg Income sources	Farming, paid labor
HH characteristics	

ID NO: 73

Interview 2

JK: Derrick

DATE: 02-11-2011
DURATION: 12.50HRS-13.50HRS;

LOCATION: BEHIND THE HOUSE, UNDER A MANGO TREE
WEATHER: IT WAS A VERY HOT AFTERNOON

Int: Do you accept to be recorded?
Res: Yes I do.

Q1 _ Int: Thank you so much. How much time has passed since your diagnosis?
Res: Its now one year and two months.

Q2 _ Int: Which factors (things, events, people, circumstances have changed the most since then?
Res: Since I got to know my status, I started accessing treatment and so I have some improvement. I no longer get sick, I had become weak but I regained my strength and I can do my work.

Int: You have said you no longer get sick. What were you suffering from?
Res: It was fever that used to disturb me.

Q28 _ Int: Have your relationship with people in your life changed since your diagnosis?
Res: I don't have any problem with people but any way people talk and those who know about it know the status.

Int: What do they say?
Res: That so and so is infected, things of the sort.

Int: Are there any changes in your daily life let me say of having fun?
Res: I married a wife that wasn't the case then. That is what has changed.

Q3 _ Int: Which things have most encouraged /promoted you your adjustment to living with HIV?
Res: I would say taking drugs and minding about taking drugs.
Int: Nothing else?
Res: Maybe even the eating habits.
Int: How? Which kind of food do you eat?
Res: This village food of ours like cassava, sweet potatoes, greens; that's if you get them.

Q4 _ Int: What is the most important thing /factor/circumstance that has helped you cope?
Res: I would think that they are those things I have mentioned. Nothing much may be to concentrate on what I do like someone can have a job and then he devotes his thoughts to it.

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